🌡️ Temperature & Humidity Sensor (AHT10)

Hardware Overview

🌡️ What is it?

The **AHT10** is a high-precision digital sensor that measures two environmental factors: Ambient Temperature (heat) and Relative Humidity (moisture in the air). It is more accurate and reliable than older analog sensors because it processes data internally before sending it to the microcontroller.

⚡ Electrical Explanation

The sensor uses a specialized capacitive element to detect moisture and a semiconductor to measure heat.

- Relative Humidity (RH): It measures how much water vapor is in the air compared to the maximum amount the air could hold at that temperature (`0%` to `100%`).
- Digital Precision: The *AHT10* converts measurements into a `24-bit` digital signal. Thus, the reading stays accurate if your wires are long or near other electronics.

Key Hardware Facts

- **Connection:** I²C Bus (`SDA`/`SCL`).
- **Default Address:** `0x38`.
- **Logic Voltage:** `3.3V` or `5V`
- **Signal Flow:** Environmental Air ➔ AHT10 Sensor ➔ ADC ➔ I²C Data ➔ Processor.

Software & Programming

📚 `kcomp_AHT10.h`

This library simplifies the 24-bit math and I²C commands into easy-to-use functions.

Usage: `#include <kcomp_AHT10.h>`

Function Reference

Function Name	Description & Parameters
`void initAHT10()`	**Setup:** Starts the I2C bus and sends a Calibration Command. **Must** be called once in `setup()`.
`float getAHT10Temperature()`	**Read Temp:** Returns the current temperature in `°C` as a decimal number.
`float getAHT10Humidity()`	**Read Humidity:** Returns the current relative humidity in `%`.
`void resetAHT10()`	**Reset:** Performs a "Soft Reset" to reboot the sensor.

⚠️ Pitfalls & Troubleshooting

- Wait for Measurement

  The AHT10 is a high-precision device that needs time to "think." Every time you request a reading, the sensor requires at least `75ms` to `100ms` to finish its internal calculations before it can send the data.

  The Fix: The library handles this with a small `delay(100)`, so avoid calling these functions in extremely high-speed loops.

- Self-Heating

  If you read the sensor constantly without any breaks, the electricity flowing through it can cause the chip to warm up slightly.

  The Fix: For the most accurate reading, only take a reading once every 2 to 5 seconds.

📝 Sample Program

This program initializes the sensor and prints the temperature and humidity to the Serial Monitor every two seconds.

```
#include <kcomp_AHT10.h>

void setup() {
    Serial.begin(115200);

    // 1. Initialize the sensor hardware
    initAHT10();
    Serial.println("AHT10 System Online!");
}

void loop() {
    // 2. Read the Temperature
    float temp = getAHT10Temperature();

    // 3. Read the Humidity
    float hum = getAHT10Humidity();

    // 4. Output results
    Serial.print("Room Temp: ");
    Serial.print(temp);
    Serial.println(" *C");

    Serial.print("Humidity:  ");
    Serial.print(hum);
    Serial.println(" %");

    Serial.println("--------------------");

    // Wait 2 seconds to prevent self-heating
    delay(2000);
}
```

💡 **Try it out:** You can find the full sample program **`TemperatureHumidityAHT10`** in the K-Comp examples folder!